Name..... Student ID .....

1. [20 points] You recorded the performance of a computer vision model for screening faulty parts in a certain factory. The results are as shown in the table below.

Actual \ Model	Passed inspection (Classified as good)	Failed inspection (Classified as bad)
Good parts	350	150
Bad parts	5	45

1.1) [4 points] What should be the positive class (Good parts or Bad parts)? Why?

1.2) [6 points] Calculate the following values.

Accuracy _____

Precision _____

Recall _____

F1_____

True Positive Rate_____

False Alarm Rate_____

1.3) [5 points] If IBM offers to sell the factory a better computer vision model. They claim that their model obtains 90% recall and 30% precision. Should you switch to use IBM's model? Why or why not? What other information might be helpful in determining this answer?

1.4) [5 points] The cost of misclassifying a good part as a bad part is 10 dollars (discarding good material). The reputation cost of selling a defective part is 100 dollars. The current model takes in an image, **x**, and output a score, **s**, that is compared against a threshold, **t**. If **s** > **t**, it is declared as failed. Otherwise, it is considered as passed. The current settings of the factory sets **t** to 3. From the provide information, is **t** set properly? Why? If not, how should we change t relative to its original value?

2. [15 points] You are starting a company to classify apples into good and bad ones. Given some data based on human experts. You try to build a machine learning model (Naïve Bayes) from this data.

Class	Color	Weight	Sweetness
Good	Green	H	T
Bad	Red	M	F
Good	Red	M	T
Bad	Green	L	T
Good	Red	M	T
Good	Green	M	F
Bad	Red	L	F
Bad	Red	L	T
Good	Red	L	F
Bad	Red	H	F

2.1) [2 points] From the data, what is the prior probability for the two classes (P(Good) and P(Bad))?

2.2) [2 points] What is P(Good| Green) and P(Bad| Green)?

2.3) [2 points] What is P(Green | Good) and P(Green | Bad)?

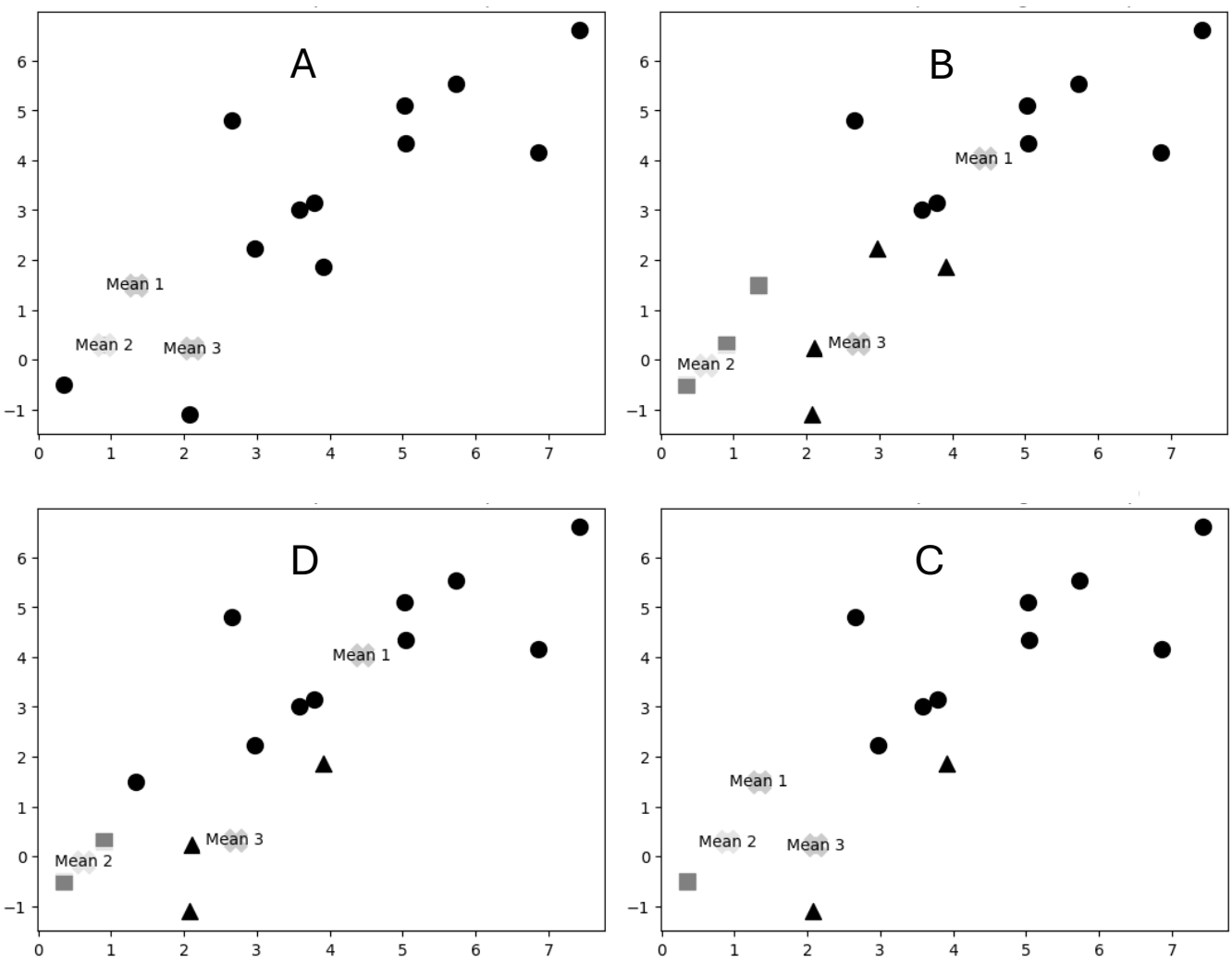
2.4) [5 points] If we do not use Naïve Bayes, what is the probability of P(Red, H, F | Bad) ? If we use Naïve Bayes (the conditional independence assumption), what is the probability of P(Red, H, F | Bad) ? Why are they different?

2.5) [2 points] If an apple is (Red, H, F), is it bad or good? Use Naïve Bayes. Show work.

2.6) [2 points] If our scale is broken, and we can only observe the color and the sweetness for the next test sample, is it bad or good? In other words, is an apple with the following features (Red, - , F) bad or good?

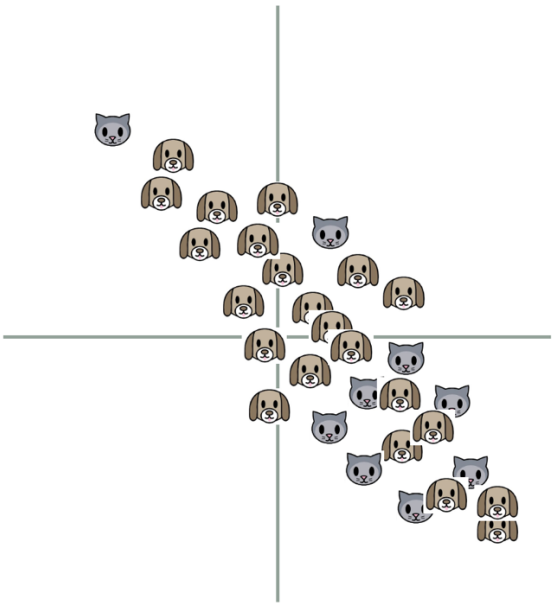
3. [5 points] The following images are pictures from the k-mean clustering algorithm. However, the images are all out of order! Put the images in the correct order so that it completes the k-mean algorithm. The first step starts with image A.

A -> _____ -> _____ -> _____



4. [10 points] Answer the following questions regarding PCA

4.1) [2 points] From the following data, draw the first PCA projection on the graph



4.2) [2 points] If the data has 1,000 dimensions, and 100 samples, how many PCA projections can there be?

4.3) [2 points] If I increase the amount of samples to 10,000, how many PCA projections can there be?

4.4) [2 points] What is the size of the covariance matrix if the data has 1,000 dimensions?

4.5) [2 points] From 4.3), what is the size of the Gram matrix? If we want to compute the PCA, should we compute from the Gram or from the covariance matrix?

5. [18 points] Consider the following neural network code.

5.1) [5 points] Draw a picture of the FeedForwardNN network. Be sure to specify the dimensions of the input/output of each layer.

```
class FeedForwardNN(nn.Module):
    def __init__(self, hidden_size=200):
        super(FeedForwardNN, self).__init__()
        self.ff1 = nn.Linear(30, 80)
        self.ff2 = nn.Linear(80, 90)
        self.ff3 = nn.Linear(90, 90)
        self.out = nn.Linear(90, 1)
    def forward(self, x):
        hd1 = F.relu(self.ff1(x))
        hd2 = F.relu(self.ff2(hd1))
        y = F.relu(self.ff3(hd2))
        y = self.out(y)
        return y
```

5.2) [3 points] How many parameters does the layer ff2 have?

5.3) [2 points] If the minibatch is of size 64, what should be the size of the input tensor?

5.4) [5 points] Draw a picture of the FeedForwardNN2 network. What should A and B be?

```
class FeedForwardNN2(nn.Module):
    def __init__(self, hidden_size=200):
        super(FeedForwardNN2, self).__init__()
        self.ff1 = nn.Linear(30, 80)
        self.ff2 = nn.Linear(A, 80)
        self.ff3 = nn.Linear(B, 90)
        self.out = nn.Linear(90, 1)
    def forward(self, x):
        hd1 = self.ff1(x)
        hd2 = self.ff2(x)
        hd3 = F.relu(hd1+hd2)
        y = F.relu(self.ff3(hd3))
        y = self.out(y)
        return y
```

5.5) [3 points] In FeedForwardNN2, the output from ff1 and ff2 are then summed together to make an output of size 80. How is this different from having just one single linear layer with 80 outputs (instead of 2)? Are they ultimately the same?

6. [12 points]

6.1) [2 points] Given an input of size 10x10x3 (HxWxC). If we apply a 2D CNN of size 3x3, stride 2, padding 1 with 5 filters, specify the size of the output tensor.

6.2) [2 points] Given a convolution neural network with connections

Input -> conv2d 3x3, stride 1 -> maxout 2x2 -> conv2d 2x2, stride 1

Find the receptive field of the final convolution layer.

6.3) [4 points] If we apply the following kernel to the input, what is the output? Assume the bias is 0, use a stride = 2, and no padding.

Input

1	2	3	0	1
4	1	0	2	3
1	3	2	1	0
0	1	4	2	1
2	3	1	0	4

Kernel

1	0	-1
0	1	0
-1	0	1

6.4) [2 points] You want to train a simple three-layer (two hidden layers) neural network for a 10-class classification. Each hidden layer has 100 neurons. The input size is 80. How many parameters are in this neural network?

6.5) [2 points] from 6.4, what are the possible hyperparameters that you can change? List at least 4 of them.